CAMPUSCYCLE

Campus Resource Sharing & Exchange Platform

REUSE • SHARE • SAVE • SUSTAIN

Updated Project Concept Report

This revised document incorporates the project's newly decided human-first moderation, reporting, review, and support-ticket workflow while preserving the original CampusCycle concept and feature structure.

Project status: Proposed system / concept under investigation

1. Project Overview

CampusCycle is a university-exclusive resource sharing and exchange platform designed around the way students acquire, use, reuse, exchange, rent, borrow, lend, and donate academic and personal resources. The central idea is not simply to create another second-hand marketplace. Instead, CampusCycle aims to connect student supply with student demand, including situations where a required resource is not currently available.

The platform is intended for resources that students commonly need for a limited period or that remain unused after a semester or academic year. Examples include textbooks, scientific calculators, laboratory coats, drawing equipment, Arduino and Raspberry Pi components, study tables, chairs, cycles, and other campus-life resources.

The concept combines a structured marketplace with borrowing, lending, renting, donation, resource requests, tag following, notifications, domain-based discovery, recommendations, and a trust-and-safety system based on verification, ratings, reporting, human moderation, and support tickets.

Important project context: This document describes the system the team is considering building. The existing campus resource-exchange system must still be studied and validated through stakeholder research, observation, surveys, and evidence before the proposed features are treated as final requirements.

Project Vision

To create a campus-specific ecosystem in which useful resources can move efficiently from students who have them to students who need them.

The system is built around five broad ideas:

- **Supply:** students can make resources available through selling, renting, lending, or donating.
- **Demand:** students can search, request, follow, or wishlist resources they need.
- **Discovery:** domains, categories, tags, search, and recommendations help users find relevant resources.
- **Matching:** the system can connect a new listing with students who have expressed interest.
- **Trust and impact:** verification, ratings, transaction history, reporting, human moderation, support tickets, and analytics support a healthier campus ecosystem.

2. Problem Statement

University students regularly acquire academic and personal resources such as textbooks, laboratory equipment, calculators, project components, electronics, and hostel essentials. Many of these resources are needed only for a particular course, semester, project, or short period of time.

The current exchange process is often fragmented across WhatsApp groups, personal contacts, hostel networks, word of mouth, notice boards, and general-purpose marketplaces. These channels can make discovery, timing, coordination, and trust difficult.

The project therefore investigates whether a centralized, university-specific system could improve the connection between available resources and unmet student demand.

The Core Problem in Simple Terms

Student A: I have an unused lab coat.
Student B: I need a lab coat, but I cannot find one.

CampusCycle is designed around creating the missing connection between these two sides.

3. Users and Stakeholders

Stakeholder	Role in the system	Typical need
Students – Buyers	Look for resources	Affordable, discoverable, trustworthy resources

Stakeholder	Role in the system	Typical need
Students – Sellers	List unused resources	Reach relevant students quickly
Borrowers	Need temporary access	Short-term, low-cost access
Lenders	Share resources temporarily	Simple lending and return process
Donors	Give resources away	Easy way to pass useful items to others
Requesters	Declare unmet demand	Be informed when an item becomes available
University Administration	Indirect stakeholder / oversight	Sustainability, safety, organized student activity
Campus businesses	Indirect ecosystem participant	Understand student resource demand
Moderators / Admins	Review reports and policy violations	Clear queues, evidence, actions, and audit history

4. Overall System Concept

Layer	Purpose	Examples
1. Supply	Represents resources students have available.	Sell, rent, lend, donate
2. Demand	Represents resources students currently want.	Search, request, follow tag, wishlist
3. Discovery	Helps users find relevant resources.	Domains, categories, tags, filters, recommendations
4. Matching	Connects demand with newly available supply.	Tag matches, request matches, notifications
5. Exchange & Trust	Supports the exchange and records its outcome.	Chat, reservation, pickup, ratings, reports, moderation, tickets, history

The most distinctive concept is the connection between **"I HAVE"** and **"I NEED"**. The platform should not become useless simply because there is no listing at the exact moment a student searches.

5. Complete Feature Set

The following list consolidates the major features discussed for the project. Some features remain core concepts, while others may be prioritized as advanced or future features after research and implementation planning.

5.1. Student Registration & University Verification

Users register as students and are verified as members of the university, preferably through an institutional email or another appropriate verification mechanism.

Verified-user badge; student profile; university-specific access.

5.2. Student Profile

Each student has a profile showing relevant activity and reputation, including listings, requests, exchange history, followed tags, wishlist, ratings, and moderation status where appropriate to the user's own account.

5.3. Buy & Sell

Students can create, edit, remove, reserve, and complete listings; set price and condition; mark items as sold; and communicate with interested buyers.

5.4. Rental System

Listings can specify rental price, duration, availability dates, optional deposit, return confirmation, and completion.

5.5. Borrow & Lend

Students can lend resources temporarily, including cases where no payment is required. The owner approves the request, the item is handed over, and both confirm its return.

5.6. Donation

Students can give resources away without selling them. Donation listings support claims and donation history.

5.7. Resource Request System

A student can declare that they need a resource even when there is no current listing. Requests can specify buy/rent/borrow, required date, duration, and status.

5.8. Domain-Based Organization

Resources are grouped into domains such as Academics, Projects & Electronics, Hostel & Campus Life, Sports & Fitness, and Other Student Resources.

5.9. Category Structure

Within a domain, resources are organized into more specific categories.

5.10. Tag System

Listings and requests can use flexible tags for discovery and matching.

5.11. Automatic / Related Tag Association

The system can associate relevant tags with a selected item or category to improve discoverability.

5.12. Follow / Adopt a Tag

Students can follow a tag when interested in a resource and receive notifications when matching resources appear.

5.13. Search & Discovery

Keyword search, tag search, domain/category browsing, and availability/price filters.

5.14. Wishlist

Students can save specific resources and receive notifications when matching listings appear.

5.15. Notifications

Notifications can cover matching listings, request matches, wishlist matches, messages, reservations, rentals, returns, and transaction updates.

5.16. Supply & Demand Matching

The system can connect listings with requests, followed tags, and wishlists.

5.17. In-App Messaging

Buyer-seller, borrower-lender, and requester-owner communication for questions, negotiation, pickup, and rental details.

5.18. Reservation

Students can temporarily reserve an item, with reservation expiry and transaction confirmation.

5.19. Campus Pickup / Exchange Locations

Predefined campus locations can support safe and convenient physical exchanges.

5.20. Ratings & Reputation

Participants can rate one another after completed exchanges; completed exchange counts and reliability indicators can support reputation.

5.21. Reporting, Human Moderation & Review

CampusCycle will use a human-first moderation workflow for the initial version. Users can report problematic content or behavior, and assigned moderators review the report before deciding what action is appropriate.

- Report a listing/post.
- Report an individual chat message.
- Report a user.
- Select a report reason such as harassment/abuse, threatening behavior, scam/fraud, prohibited content/item, spam, inappropriate content, or other.
- Allow an optional explanation from the reporter.
- For chat-message reports, retain the reported message and relevant surrounding conversation context for moderator review.
- Moderation queue with Pending, Under Review, Resolved, and Dismissed states.
- Moderator actions: dismiss, warn, remove content, restrict features, temporarily suspend, or permanently suspend.
- Record moderation decisions and actions in an audit/history trail.
- Do not automatically punish a user solely because a word appears in a message; context and the surrounding conversation matter.

Example: “I’ll find you after class” may be normal or threatening depending on the conversation. The moderator reviews the context rather than relying on a keyword alone.

5.22. Support Ticket System

Reports handle specific pieces of content or behavior. A separate ticket system handles broader problems that require investigation or follow-up.

- Transaction/payment or delivery disputes.
- Account problems.
- Appeals against moderation actions.
- Repeated harassment or problems involving multiple messages/listings.
- General complaints or issues that do not map cleanly to one report.

Tickets can include a category, user description, attachments/evidence where appropriate, status, assigned moderator/support person, conversation history, and final resolution.

5.23. Admin / Moderator Dashboard

Administrators and assigned moderators have tools for managing users, listings, reports, tickets, categories, domains, tags, and platform activity.

- View and prioritize pending reports.
- Open the reported content and relevant context.
- Review and resolve reports.
- Manage support tickets.
- Warn, restrict, remove, or suspend accounts/content.
- View moderation history and previous actions.
- Manage users, categories, domains, and tags.

5.24. Recommendations

When a student views a resource, the platform can recommend related resources based on activity, category, domain, or resource relationships.

5.25. Resource Ecosystems

Some resources naturally belong together. CampusCycle can represent these relationships so students discover groups of useful resources rather than isolated items.

5.26. Demand–Supply Analytics

The system can compare how many students are looking for a resource with how many matching resources are available.

5.27. Sustainability & Impact Dashboard

The platform can measure validated reuse, donation, exchange, and student-saving outcomes rather than relying on invented impact numbers.

5.28. Transaction & Activity History

Students can see past purchases, sales, rentals, borrowing, lending, donations, fulfilled requests, and other relevant activity.

5.29. Status Tracking

Listings and requests move through clear states such as Available, Reserved, Sold, Rented, Borrowed, Donated, Unavailable, Active, Matched, Fulfilled, Expired, and Cancelled.

6. Key Workflows Explained

Workflow A — Searching and Following a Tag

Search #coat → No listing → Follow #coat → New listing appears → Match → Notification

Workflow B — Student Creates an Explicit Request

Student specifies the item, transaction type, required date, and optional duration. The request remains active until fulfilled, cancelled, or expired and can be matched to future listings.

Workflow C — Listing an Item

Student selects a domain/category, adds photos, description, condition, price or rental terms, availability, and tags. The listing becomes searchable and eligible for matching.

Workflow D — Viewing a Resource and Getting Recommendations

The system can show related resources based on domain, category, or resource relationships.

Workflow E — Complete Exchange

Browse/search → Open listing → Check details → Chat → Negotiate/confirm → Reserve → Meet at campus pickup location → Complete exchange → Update status → Rate participant → Save transaction history.

Workflow F — Report & Moderator Review

Step 1: A user sees a problematic listing, user, or chat message.
Step 2: They select Report and choose a reason, with an optional explanation.
Step 3: The system creates a report and records the relevant content; message reports include surrounding context where appropriate.
Step 4: The report enters the moderator queue as Pending.
Step 5: An assigned moderator opens it and moves it to Under Review.
Step 6: The moderator evaluates the content and context.
Step 7: The moderator either dismisses the report or takes an appropriate action such as warning, content removal, restriction, or suspension.
Step 8: The system records the decision in moderation history and notifies the relevant user where appropriate.

Workflow G — Support Ticket

User creates ticket → selects category → describes problem → attaches evidence if needed → ticket enters queue → moderator/support person investigates → additional information requested if needed → decision/resolution recorded → ticket closed.

7. Example Domain Structure

Domain	Example categories/resources
Academics	Books, calculators, laboratory equipment, lab coats, drawing materials
Projects & Electronics	Arduino, Raspberry Pi, sensors, breadboards, tools, cables
Hostel & Campus Life	Study tables, chairs, lamps, extension boards, room essentials
Sports & Fitness	Cycles, sports equipment, fitness accessories
Other Student Resources	Stationery, notes, printouts, event equipment, miscellaneous resources

Hierarchy example: Projects & Electronics → Arduino → Arduino Uno → #arduino, #electronics, #project, #embedded.

8. Supply–Demand Concept

SUPPLY — “I HAVE IT”

- List an item for sale, rent, lending, or donation.
- Specify availability, condition, location, and transaction terms.
- Become discoverable through domains, categories, tags, and search.

DEMAND — “I NEED IT”

- Search for a resource.
- Follow a tag when the resource is currently unavailable.
- Create a specific request.
- Save a resource to a wishlist.

The matching idea: When supply appears later, the system can connect it to previously expressed demand.

9. Trust, Safety & Moderation

Because the platform involves student-to-student exchanges, trust is an important part of the system. CampusCycle therefore uses several mechanisms rather than assuming every participant is automatically trustworthy.

- **University verification:** restrict participation to verified students where feasible.
- **Profiles:** provide context about a participant's activity.
- **Ratings:** allow participants to rate completed exchanges.
- **Transaction history:** provide a record of completed activity.
- **Reporting:** allow users to report suspicious, abusive, misleading, inappropriate, or otherwise problematic content and users.
- **Human moderation:** assigned moderators review reports and decide appropriate actions using content and context.
- **Support tickets:** provide a structured channel for broader disputes, account issues, appeals, and problems requiring investigation.
- **Audit history:** record moderation decisions, actions, and timestamps for accountability.

Initial moderation principle

CampusCycle will not rely on automatic punishment based only on restricted words. A message can have different meanings depending on its context. The initial system therefore emphasizes user reporting and human review. AI-assisted moderation may be considered later as a pre-screening or prioritization tool, not as the foundation of the MVP.

10. Analytics & Impact

Metric	What it could show
Resources listed	How many resources students make available
Resources reused/exchanged	How many listings lead to successful exchanges
Requests fulfilled	How often expressed demand finds a matching resource
Demand vs supply	Resources with high demand but limited supply
Student savings	Estimated money saved where transaction data supports the calculation
Donation activity	Resources passed on without a sale
Moderation activity	Report volume, resolution status, action types, and recurring categories of policy violations
Support activity	Ticket volume, categories, resolution status, and response/resolution patterns

A future sustainability dashboard could summarize validated measures such as resources reused, donated items, successful exchanges, student savings, and other defensible impact measures.

11. What Makes the Concept Different?

Capability	Typical general marketplace	CampusCycle concept
General buying/selling	Yes	Yes
University-only community	Usually no	Yes
Renting resources	Limited / varies	Core use case
Borrow/lend	Usually informal	Structured
Donation	Usually separate/informal	Integrated
Explicit unmet-demand requests	Limited	Core concept
Follow a resource tag before supply exists	Uncommon	Core concept
Campus-specific domains	No	Yes
Resource recommendations	Common in e-commerce	Adapted to academic/resource ecosystems
Demand–supply analytics	Not campus-specific	Possible
Human-first reporting & moderation	Varies	Integrated
Support tickets / appeals	Varies	Integrated

The strongest differentiator is not the ability to sell a second-hand item. It is the attempt to connect unmet student demand with future or existing campus supply through tags, requests, matching, notifications, and domain-aware discovery.

12. High-Level System Architecture

A possible technical architecture is shown below. This is a conceptual architecture and can be simplified or changed after the team finalizes requirements.

Student / Admin / Moderator → Web Frontend → Backend API → Database
■ File/Image Storage
■ Notification Service
■ Search / Matching / Recommendation Logic
■ Reporting / Moderation / Ticket Service

Possible technology choices:

- Frontend: React with a utility-first CSS framework such as Tailwind CSS.
- Backend: Node.js and Express.
- Database: MongoDB or another suitable database selected after schema analysis.
- Authentication: institutional email verification and/or a managed authentication provider.
- Image storage: a managed object/file storage service.
- Deployment: a simple cloud deployment suitable for a student project.

These technologies are examples, not final requirements. The systems-thinking work should remain focused on understanding the problem and system before technology decisions become fixed.

13. Suggested Feature Prioritization

Priority	Features
Core / MVP	Student registration and verification; profiles; domains/categories; listings; buy/sell; rent; borrow/lend; donate; search; tags; requests; follow tags; notifications; messaging; reservations; ratings; reporting; human moderator review; moderation actions; moderation history; admin/moderator dashboard; support tickets.
Advanced	Wishlist; automatic tag association; supply-demand matching; recommendations; resource ecosystems; campus pickup workflows; demand/supply analytics; sustainability dashboard.
Future / Research-dependent	Personalized recommendation engine; machine-learning demand prediction; image-based categorization; AI-assisted moderation pre-screening; inter-college exchange; advanced impact estimation; mobile application.

14. Complete Example: The Lab Coat Scenario

A student needs a lab coat for two weeks, but there is currently no lab coat available.

1. The student searches #coat.
2. The system searches related tags such as #labcoat, #laboratory, and #labequipment.
3. No suitable listing is available.
4. The student chooses Follow #coat or creates an explicit request.
5. The system stores the student's demand.
6. Three days later, another student lists a Lab Coat for rent.
7. The listing receives relevant tags automatically or through the selected category.
8. The matching system detects that the new listing is relevant to the earlier demand.
9. The student receives a notification.
10. The student opens the listing, chats with the owner, and reserves the coat.
11. The student meets the owner at a campus pickup location and completes the rental.
12. After the return, both parties confirm completion and can rate the exchange.
13. While viewing the listing, the system may recommend related laboratory resources.

14. If a participant encounters abusive or suspicious behavior during the exchange, they can report the relevant message/listing/user for moderator review or open a support ticket for a broader issue.

What this demonstrates: Domains + tags + demand + follow/adopt + matching + notifications + rental + messaging + reservation + campus exchange + ratings + recommendations + reporting/moderation + support.

15. Current Project Position & Next Steps

The project is currently being developed as a systems-thinking study. The team has considered the campus resource exchange problem, stakeholder services, and a proposed concept. The next step is to validate assumptions through evidence from the actual university environment.

- Map the current As-Is resource exchange process.
- Identify and validate stakeholder needs.
- Conduct student surveys and interviews.
- Collect quantitative evidence for important claims.
- Identify symptoms, patterns, structures, and possible root causes.
- Determine which proposed features actually address validated problems.
- Prioritize the final requirements before implementation.
- Build the smallest useful prototype after the system study is sufficiently validated.

Important: CampusCycle should not be treated as a proven solution merely because the team has designed an appealing feature set. The systems-thinking phase should establish which problems genuinely exist, how frequently they occur, who is affected, and why they occur. The final solution should then be refined around that evidence.

16. One-Page Summary

Area	CampusCycle concept
Problem	Students who have reusable resources and students who need them may fail to find each other through fragmented exchange channels.
Users	University students, plus administration, moderators, and other indirect ecosystem stakeholders.
Supply	Sell, rent, lend, donate.
Demand	Search, request, follow tags, wishlist.
Discovery	Domains, categories, tags, keyword search, filters, recommendations.
Matching	Connect listings with requests, followed tags, and wishlists; notify interested students.
Exchange	Chat, negotiate, reserve, campus pickup, complete transaction, rate.
Trust	University verification, profiles, ratings, reports, human moderation, support tickets, history.
Insights	Demand/supply analytics, resource reuse, donations, exchanges, student savings, moderation/support trends.
Differentiator	The platform exposes both supply and unmet demand and connects them over time.

Core idea to remember: CampusCycle is not just a place where students sell second-hand products. It is a proposed campus resource ecosystem that tries to make "I HAVE" and "I NEED" visible to each other, even when the match does not exist at the same moment.